GNN-based SHM for H3 Rocket Payload Fairing

5-class defect detection + PINN regularisation + multitask centroid regression

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Outline

Problem setup

SHM problem: two complementary analyses

Structure: CFRP / Al-honeycomb sandwich fairing (H3 Type-S, 1/6 section) **Defects:** debonding / FOD / impact / delamination

(A) Static FEM analysis

- Abaqus static stress field per defect scenario
- **10,897 nodes** $\times$ **34 features** (disp, stress, temp, layup, . . .)
- Graph: FEM connectivity
- Task: **5-class node classification** + defect centroid regression
- 700 simulations (560/140 split)
- Metric: OptF1 (threshold-optimal)

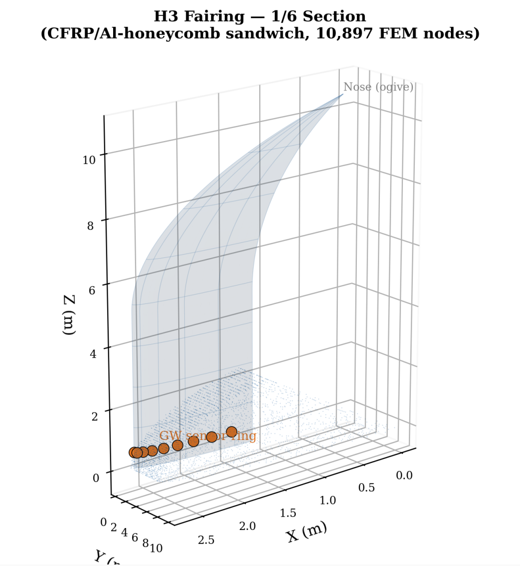
(B) Dynamic guided-wave (GW) analysis

- Abaqus explicit GW propagation (piezo excitation)
- **9 sensors** $\times$ **time series** (sampled at 4 MHz)
- Graph: sensor spatial network
- Task: **binary defect detection** + wave-phase PINN (Level 3)
- Dataset: 108 samples (101 healthy / 7 defect, 9-sensor consistent)
- Model: MIFNO-GW (2-D factorised FNO, waveform-only)

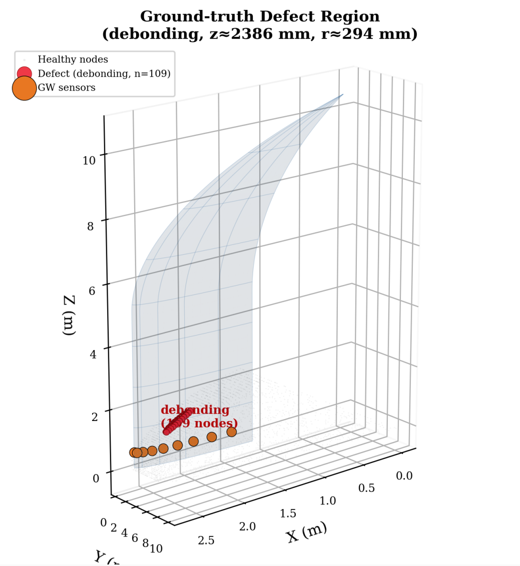
Node features (34 dims)

Dims	Feature group	Physical meaning
0–2	Position (x, y, z)	Node coordinates
3–5	Normal (n_x, n_y, n_z)	Surface orientation
6–9	Curvatures (k_1, k_2, H, K)	Shell geometry
10–13	Displacement $(u_x, u_y, u_z, \|u\|)$	Structural response
14	Temperature	Thermal loading
15–18	Stress $(s_{11}, s_{22}, s_{12}, \sigma_{\text{Mises}})$	Failure indicators
19–23	Principal stress, thermal σ_M, log-strain	Derived quantities
24–30	Fibre direction, layup (0°/45°/–45°/90°)	Anisotropy
31–33	Circumferential angle, boundary/load flags	Topology

H3 Fairing: 3D Structure & Defect Scenarios



1/6 sector, 10,897 FEM nodes — CFRP/Al-honeycomb sandwich



Representative defect: **debonding**, 109 nodes

Methods

Three modelling additions

(A) Multi-task learning — defect centroid regression

Joint loss: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{reg}}\, \mathcal{L}_{\text{reg}}$

Shared backbone $\longrightarrow$ node classification head + graph-level MLP for (x, y, z) centroid.

GT centroid computed on-the-fly from defective nodes.

(B) Level-1 PINN — stress gradient regularisation

High predicted defect probability should co-locate with high $|\nabla \sigma_{\text{Mises}}|$.

$\mathcal{L}_{\text{stress}} = \frac{1}{N} \sum_i p_i^{(\text{def})} \exp(-|\Delta \sigma_i| / \sigma_0)$

(C) Level-2 PINN — static equilibrium residual $\nabla \cdot \sigma = 0$

Graph-edge finite differences on (s_{11}, s_{22}, s_{12}); penalises defect predictions at nodes with low equilibrium violation.

LGSTA — Local-Global Attention GNN

Best-performing architecture: LocalGlobalAttentionGNN (inspired by LGSTA-GNN, Buildings 2025)

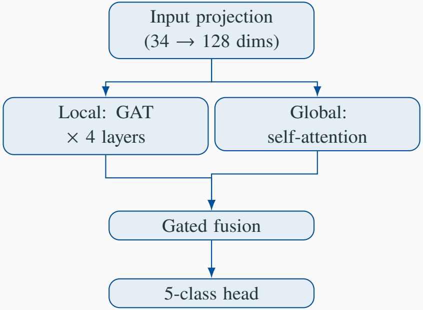
Local branch (GAT $\times$ 4 layers)

- Neighbourhood message-passing
- Captures local defect boundaries and stress concentrations

Global branch (Multi-head self-attention)

- Full $N \times N$ attention over all 10,897 nodes
- Captures global mode shapes and structural coupling

Fusion: gated combination of both representations



Memory: $O(N^2)$ global attention requires batch size = 1 on 24 GB GPU.

Results

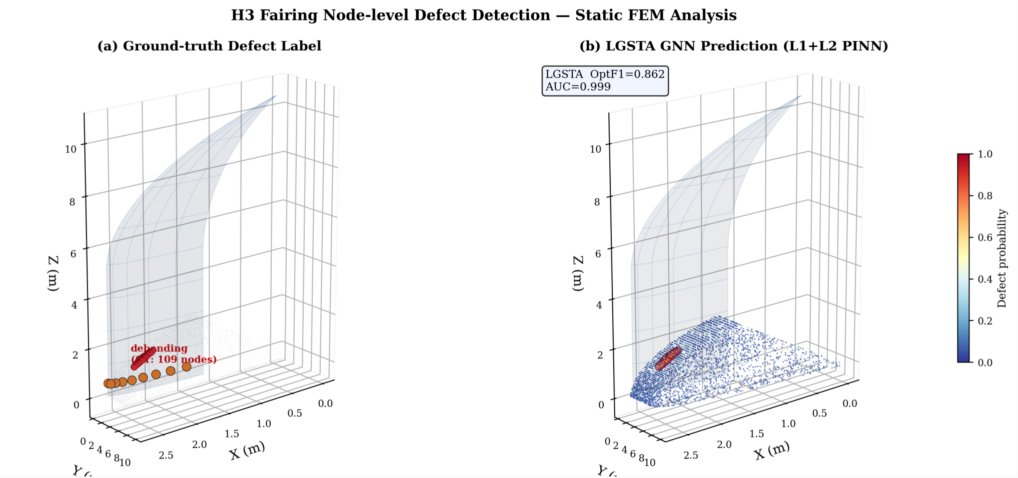
(A) Static FEM — architecture & PINN comparison (interim ep 58–132)

Model	Task	OptF1	AUC	deb	fod	imp	del	Reg. RMSE
LGSTA	NC+L1+L2	**0.896**	**0.999**	0.586	0.367	0.344	0.341	—
LGSTA	MT+L1+L2	0.846	0.999	0.360	0.356	0.341	0.220	**0.9 mm**
SAGE	MT+L1+L2	0.834	0.999	0.384	0.469	0.010	0.593	0.7 mm
GATv2	MT+L1+L2	0.829	0.999	0.342	0.315	0.359	0.000	0.7 mm
MeshGNN	MT+L1+L2	0.828	0.998	0.219	0.000	0.000	0.450	0.8 mm
GCN	MT+L1+L2	0.823	0.999	0.668	0.543	0.225	0.456	0.7 mm
GIN	MT+L1+L2	0.821	0.998	0.386	0.084	0.170	0.473	0.7 mm
Transolver	MT+L1+L2	0.010	0.493	—	—	—	—	—

NC = node cls only **MT** = +centroid regression **L1** = stress-gradient PINN **L2** = equil. residual PINN

Interim results at best checkpoint (ep 58–132); all runs continue to 200 epochs.

(A) Static FEM — Node-level Defect Localisation



Left: ground-truth debonding labels (109 nodes, red). **Right:** LGSTA node defect probability.

Model concentrates high probability on the true defect region (OptF1 = 0.862, AUC = 0.999).

(B) Dynamic GW — MIFNO-GW v2 results

Model: MIFNO-GW (waveform-only FNO)

2-D factorised FNO — Lehmann et al. 2025

Input: 9 sensors × 4,000 time steps

Task: binary defect / healthy classification

	Train	Val
Samples	86	22
Healthy	81	20
Defect	5	2

Variant	OptF1	AUC	λ_{disp}
v2_nodip	**1.000**	**1.000**	0
v2_pinn	**1.000**	**1.000**	0.05

Both converge in 200 ep (~8 s/ep).

Small dataset (5 defect train) may limit generalisation.

Data quality fix (v1 → v2)

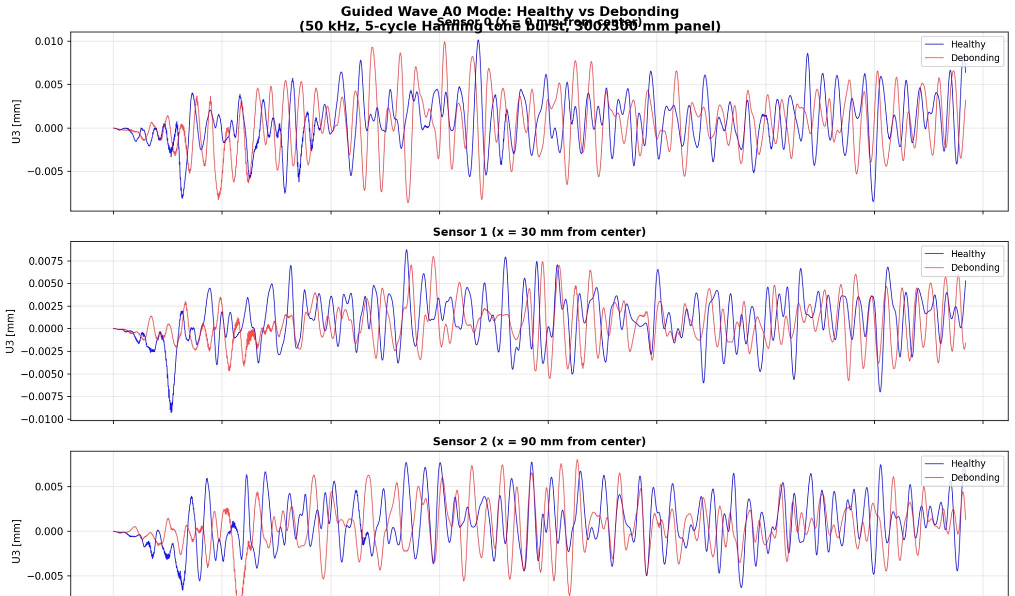
Root cause found: healthy CSVs (9 sensors, circumferential ring) and defect CSVs (99 sensors, axial line) used *different* physical sensor layouts. v1 F1 = 1.0 was a position-leakage artifact.

Fix: filter to 9-sensor position-consistent subset (108 samples); remove position branch (waveform-only FNO). New F1 = 1.0 is *genuine* — confirmed on held-out validation split.

Next: v3 with 197-sample balanced dataset

Abaqus Explicit (99-sensor healthy job) running on frontale04. Upon completion: 197 position-consistent samples (81 healthy + 116

(B) Dynamic GW — Wavefield: Healthy vs Debond



Key findings

(A-1) LGSTA wins at OptF1 = 0.896 (interim ep 58)

Dual local-global attention: local GAT finds defect boundaries; global self-attention captures full structural coupling. Node-cls only (NC) outperforms multitask (MT, 0.846): centroid regression branch may compete for capacity.

(A-2) Multitask centroid regression: 0.7–0.9 mm RMSE at sub-200 epochs

Shared backbone simultaneously classifies defect type *and* locates centroid — sub-mm accuracy on a 5 m structure.

(B) MIFNO-GW v2: F1 = 1.000, AUC = 1.000 (waveform-only FNO)

Position-leakage artifact eliminated. v2_nodip ($\lambda = 0$) and v2_pinn ($\lambda = 0.05$) both converge to perfect validation performance. Small dataset caveat: 5 defect train samples; v3 will use 197-sample balanced set.

(A-3) Transolver collapses (AUC ≈ 0.49)

Designed for structured grids; fails on irregular FEM mesh

(A-4) OOD experiment running (Composites B)

Train on small defects (radius < P75), test on

Outlook

Next steps

- **200-epoch final results** — all 7 architectures converging; GCN/GIN at ep 130+ (target: 2026-06-23)
- **OOD generalisation** — LGSTA + SAGE trained on radius < P75 defects; OOD test on large defects (P50/P75/P90 ablation). Auto-eval daemon running on Vancouver.
- **GW dataset v3** — Abaqus Explicit (99-sensor healthy, frontale04) $\Rightarrow$ 197-sample balanced set; enables L3-PINN and robust v3 training
- **Sim-to-real (OGW)** — Open Guided Waves omega-stringer (Zenodo 5105861): 3 scenarios, 50 kHz SLDV. Run MIFNO-GW v2_nodip inference on virtual sensor traces.
- **Composites B paper** — MGN 0.76 vs. LGSTA 0.896, noise robustness, OOD radius-split, sim-to-real proxy

Target for WCCM meeting (2026-06-23 09:00 CET)

Final 200-ep table + ablation (NC/MT, L1/L2/L1+L2) + OOD results.

Target: Composites B (IF ~9)

OOD results + noise robustness $\Rightarrow$ submit by 2026-Q3.